Fiscal Year:	FY 2024	**Task Last Updated:**	FY 09/26/2024
PI Name:	Smith, Amanda L. Ph.D.		
Project Title:	Interface Consistency (Phase I): Characterization and Measurement		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **HSIA**:Risk of Adverse Outcomes Due to Inadequate Human Systems Integration Architecture		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	amanda.l.smith-1@nasa.gov	**Fax:**	FY
PI Organization Type:	GOVERNMENT	**Phone:**	
Organization Name:	KBR/NASA Johnson Space Center		
PI Address 1:	2101 Nasa Pkwy		
PI Address 2:			
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	36
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	12/05/2022	**End Date:**	12/05/2023
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
Contact Email:	alexandra.m.whitmire@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Holden, Kritina Ph.D. (Leidos)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Future exploration missions will rely on designing and developing vehicles and complex systems from within NASA and through multiple external commercial partners to meet mission goals. Despite existing consistency-related agency requirements, NASA’s approach to commercial spaceflight development encourages providers’ flexibility and innovation. This strategy is resulting in significant design diversity across Artemis vehicles. Design best practices and guidelines champion interface consistency to promote mental model development and knowledge transfer. However, research investigating the benefits of consistency is mixed, and little is known about its role in complex systems. Determining the level of risk that system diversity presents is difficult, as there is no established method for quantifying the degree of consistency within and across interfaces, nor is there information about the differential impacts of different types of inconsistency.		

Rationale for HRP Directed Research:	Future crewed exploration missions beyond low Earth orbit (LEO), including Artemis, will be made possible by many development efforts from both within NASA and by external partners. The number of different vehicles and systems requiring human interaction will be much greater for future missions. Design solutions across programs and projects are likely to vary, and with NASA's current approach of allowing commercial providers great flexibility in design, the risk of inconsistency is significant. Artemis development efforts are already in progress, which underscores the urgency of this work. This proposal qualifies for directed study due to the insufficient time required for solicitation. Another factor is the highly constrained nature of the work. The proposers are currently involved in commercial programs as subject matter experts and are knowledgeable about the diversity of designs currently existing, and the development processes and constraints under which consistency measures and standards would have to be implemented. External researchers would not easily have access to this information; thus, directed research would be more cost effective.
Research Impact/Earth Benefits:	
Task Progress:	Phase I of this project served as a starting point to better understand the construct of consistency, its application, and the range of studies and methods for measuring it. The project team created a taxonomy of consistency to apply to interfaces as a framework to guide the development of tools to assess intersystem consistency. Checklist and cognitive walkthrough methods were developed for use by human factors (HF) and human-computer interaction (HCI) experts. The Intersystem Consistency Scale (ICS) was developed for interface evaluations with crew. A pilot study evaluated the methods' ability to distinguish differences between Artemis-like prototype pairs exhibiting either high or low design consistency. In addition, click errors and time on task were collected within the ICS (crew-like) group. Results from our exploratory analysis suggest each of the methods can discriminate between prototype pairs of high and low consistency and that low consistency is associated with reduced efficiency and increased workload.
Bibliography Type:	Description: (Last Updated: 04/28/2026)